

Retail Sales Data Quality & Business Analysis

Excel | Power Query | MySQL | Power BI

Transactions	Sales	Profit	Margin
200	\$206,813.95	\$13,042.51	6.31%

1. Project Overview

A retail sales analytics project focused on validating transaction and customer data, building a repeatable Excel reporting workflow, investigating business performance, and communicating findings through dashboards.

2. Business Problem

The goal was to determine whether a retail transaction dataset was reliable enough for analysis and to turn the validated data into useful sales, profitability, customer, and regional insights. The analysis also needed to identify inconsistencies that could affect customer reporting and decision-making.

3. Data Quality & Validation

The dataset contained 200 transaction records. Eight core validation checks passed. A deeper customer-identity check found 85 unique Customer IDs and 199 unique customer names. 61 of the 85 Customer IDs (71.76%) were associated with multiple names. This issue was flagged for review rather than manually corrected because the data did not provide enough evidence to determine the correct identity.

4. Excel System & Power Query Workflow

Power Query was used to clean and standardize the source dataset before analysis. An Operations Summary was built using XLOOKUP and SUMIFS to connect Product IDs with product names and calculate product-level quantity, sales, profit, and margin. Reconciliation controls compared the summary with the source transaction data; sales and quantity differences were both zero.

5. Business Analysis

The analysis produced \$206,813.95 in total sales, \$13,042.51 in total profit, a 6.31% overall profit margin, 200 orders, 1,115 units sold, and a \$1,034.07 average order value. There were 61 repeat customers and 24 one-time customers.

6. Regional & Product Investigation

South generated the highest regional sales at \$55,317.47, while Central produced the highest regional profit margin at 11.97%. East generated \$53,139.57 in sales but had the lowest margin at 1.81%. A drill-down from region to category to product identified Technology in East as loss-making and highlighted Accessories Product 13 as a significant loss contributor, with \$3,561.53 in sales and -\$325.48 in profit. This was treated as an investigation finding rather than a definitive causal explanation.

7. Dashboards & Reporting

Excel was used for validation, structured analysis, PivotTables, lookups, reconciliation, and dashboard reporting. Power BI was used to communicate KPIs, sales trends, category and segment mix, and regional performance through an interactive dashboard.

8. Key Outcomes

The project converted a raw 200-row dataset into a validated, reconciled, and documented reporting workflow. It surfaced a material customer-data integrity issue, quantified business performance, and demonstrated a repeatable approach for moving from data quality checks to business insight.

9. Tools & Skills Demonstrated

Excel: Power Query, XLOOKUP, SUMIFS, PivotTables, data validation, reconciliation, dashboarding.

MySQL: data-quality checks, aggregation, customer analysis, profitability analysis, trend analysis.

Power BI: KPI reporting, interactive filters, regional/category visualization, and trend reporting.

10. Limitations & Next Steps

The customer-name inconsistency was identified and quantified but not manually corrected because the dataset did not contain sufficient evidence for authoritative customer resolution. Future enhancements could include a formal customer master table, stronger data-governance rules, automated exception reporting, and additional Python-based analysis as part of the broader portfolio.